Facultad de Ciencias
UNAM

AVANCES DE TESIS DE LICENCIATURA

Cálculo numérico de propiedades ópticas de eritrocitos sanos y enfermos

Dana Larissa Luna González

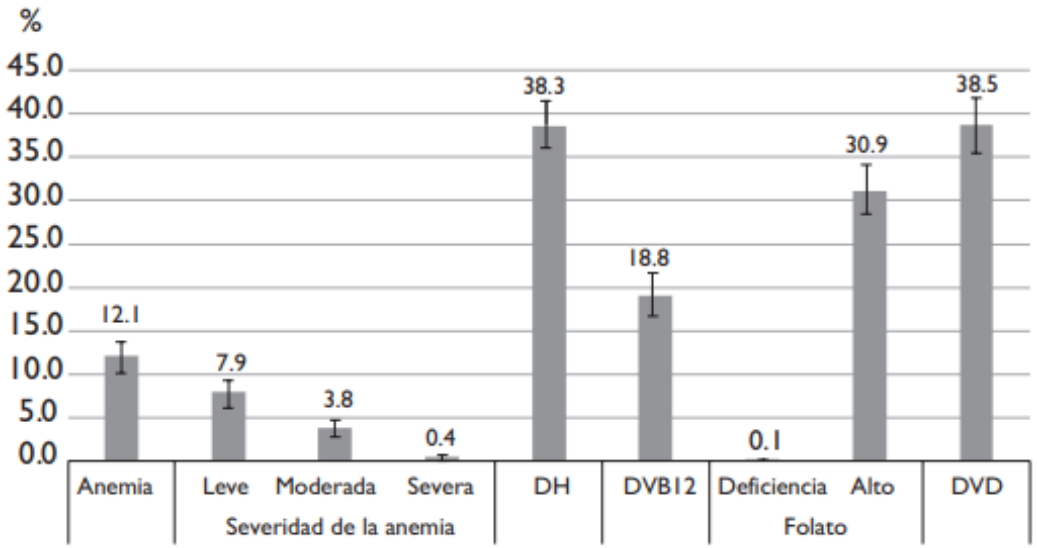
M. en C. Jonathan Alexis Urrutia Anguiano
Dr. Alejandro Reyes Coronado

Motivación

Anemia en México

- En México, la prevalencia de anemia en los últimos 4 años alcanza el 12.1%

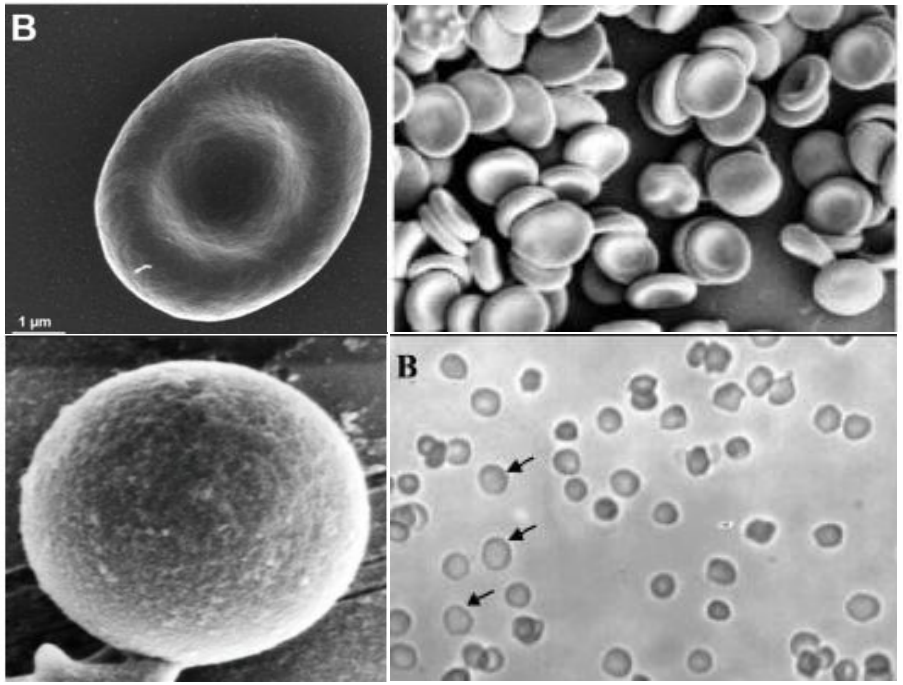
- La anemia es una disminución de la capacidad de transporte de oxígeno de la sangre. Se clasifica según:
  - Morfología
  - Composición

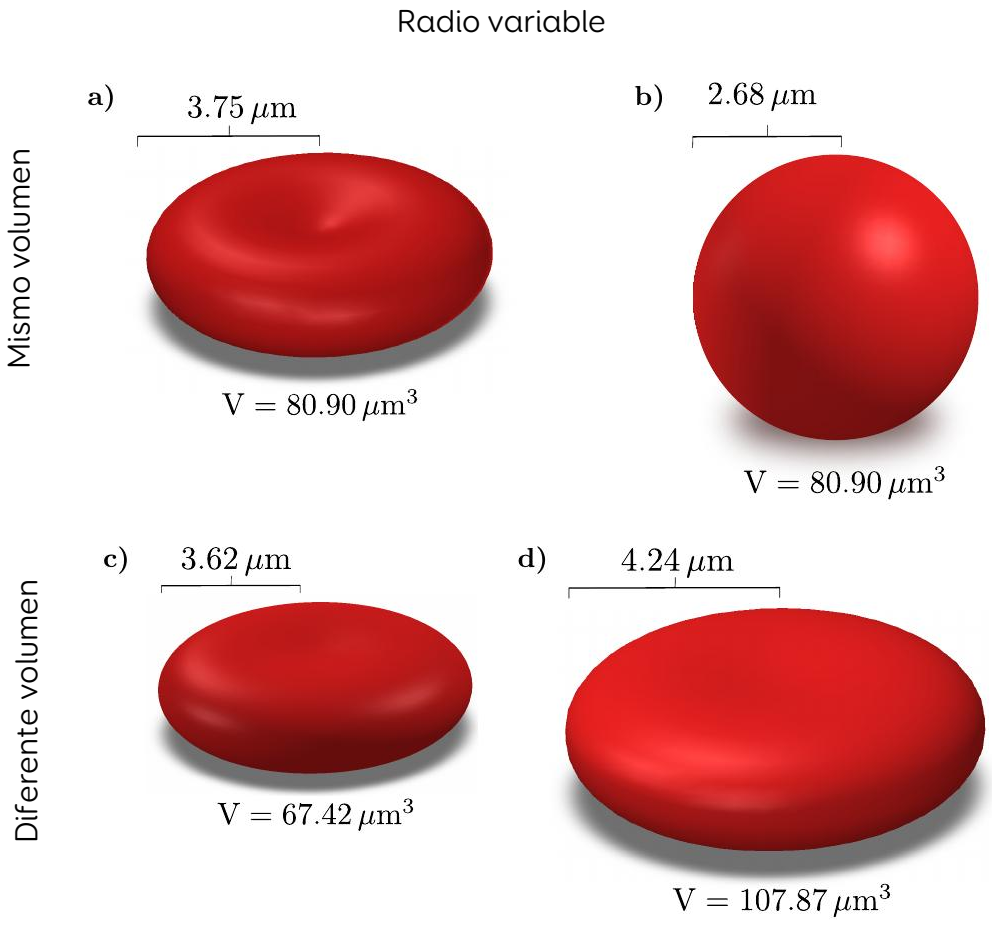


Shamah-Levy, T., et al. *Salud Pública de México*, **67**(6), 2025

Motivación

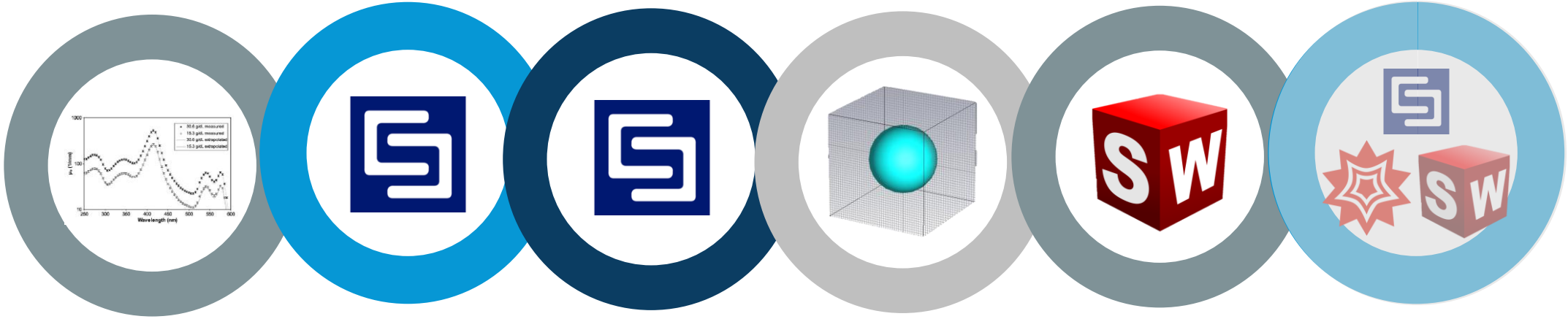
Morfología de eritrocitos sanos y enfermos





Alummoottil, M. , et al. *Hemato*, **15**(4), 2010
Khanna, R., et al. *Proceedings of the National Academy of Sciences*, **99**(10), 2002

Plan de trabajo



- Sano
- Microcitosis
- Macrocitosis
- Esferocitosis hereditaria

Construcción de funciones dieléctricas de eritrocitos

Comparación de CST vs Mie

Análisis de convergencia numérica en CST

Evaluación del efecto del mallado

Construcción de geometrías de eritrocitos

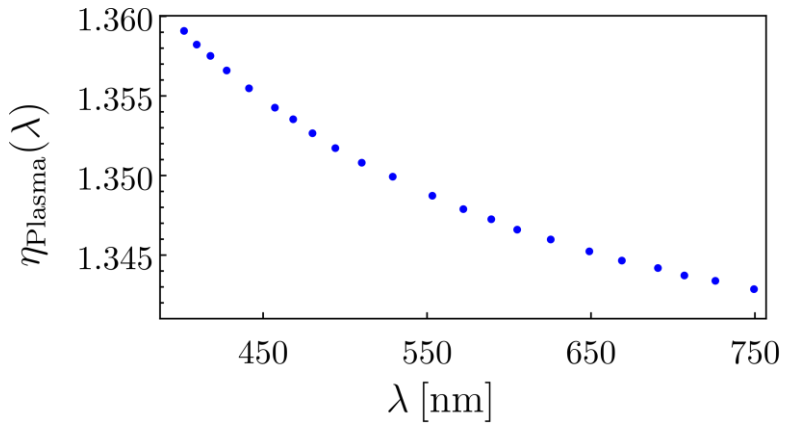
Cálculo de propiedades ópticas de eritrocitos

Redacción → Teoría → Resultados y conclusiones

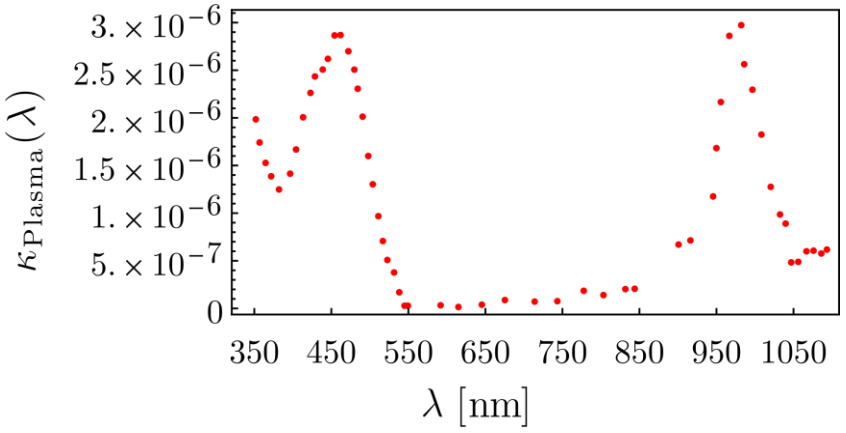
Datos experimentales

$$n(\lambda) = \eta(\lambda) + i\kappa(\lambda)$$

Plasma



Liu, S., et al. *Journal of Biomedical Optics*, **24**(3), 2019



Meinke, M., et al. *Journal of Biomedical Optics*, **12**(1), 2007

Eritrocitos

Rango espectral

Parte real: 250 - 1100 nm

$$\eta_{\text{Hb}}(\lambda, c_{\text{Hb}}) = \eta_{\text{H}_2\text{O}}(\lambda)[\beta(\lambda)c_{\text{Hb}} + 1]$$

Friebel, M. y Meinke, M. *Applied Optics*, **45**(12), 2006

Parte imaginaria:

Concentración	$\kappa_{\text{Hb}}(\lambda)$	Rango espectral
• 28.7 g/dL	Microcitosis	250 - 600 nm
• 30.6 g/dL	Sano	250 - 1050 nm
• 34 g/dL	Macrocitosis/esferocitosis	250 - 600 nm

Friebel, M. y Meinke, M. *Applied Optics*, **10**(6), 2005

Funciones dieléctricas

1. Obtención de la parte real e imaginaria de la función dieléctrica mediante las relaciones

$$\varepsilon' = \eta^2 - \kappa^2$$
$$\varepsilon'' = 2\eta\kappa$$

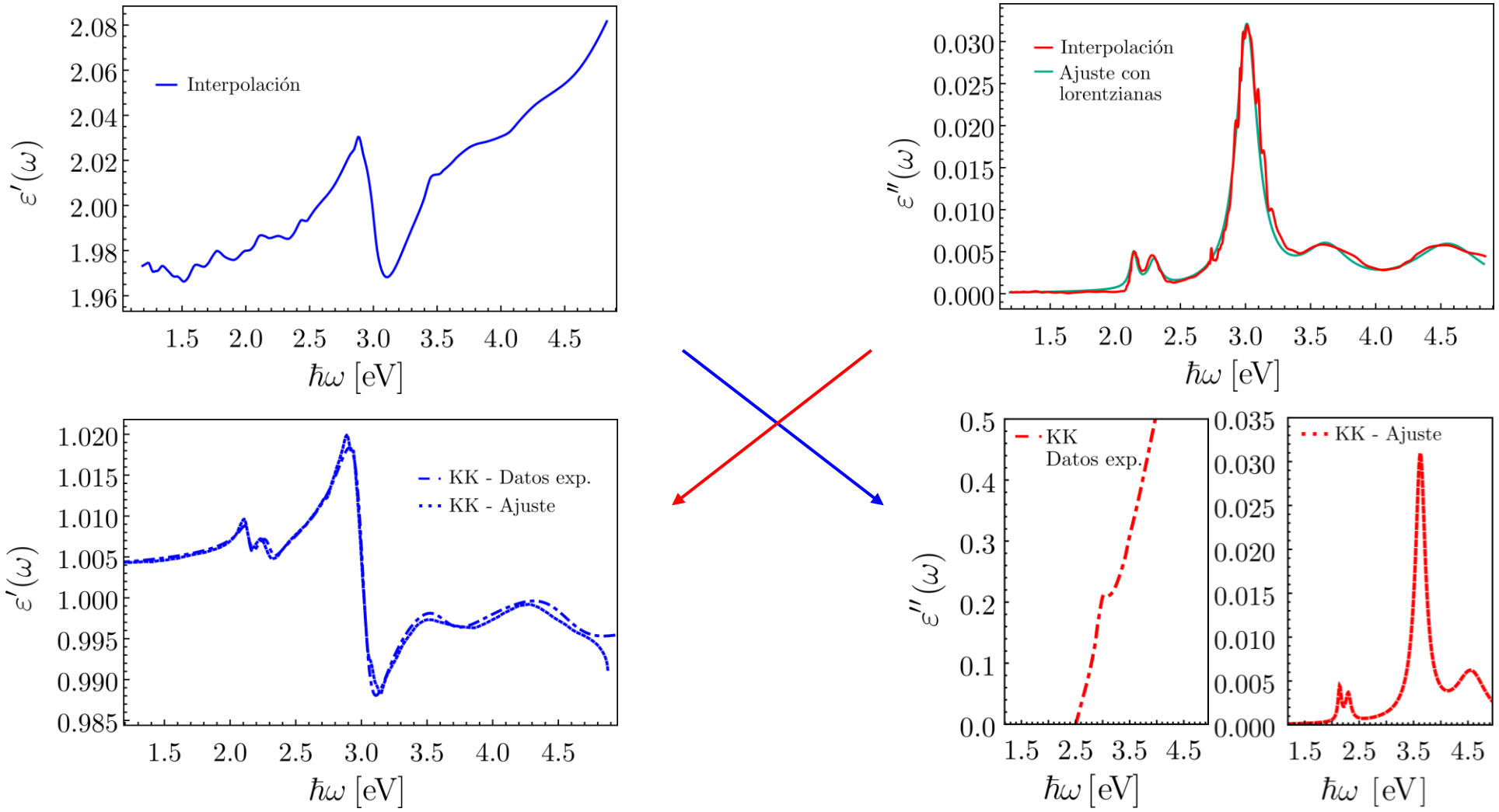
2. Ajuste de lorentzianas a la parte imaginaria de la función dieléctrica

$$\varepsilon'' = \sum_{i=1}^{6} \frac{A_i \gamma_i \omega}{(\omega_{0_i}^2 - \omega^2)^2 + \gamma_i^2 \omega^2}$$

3. Aplicación de las relaciones de Kramers-Kronig (KK) y las relaciones sustractivas de KK (SSKK)

$$\varepsilon'(\omega) = \varepsilon'(\omega_1) + \frac{2(\omega^2 - \omega_1^2)}{\pi} \mathrm{PV} \int_0^\infty \frac{\omega' \varepsilon''(\omega')}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} \mathrm{d}\omega'$$
$$\varepsilon''(\omega) = \varepsilon''(\omega_1) - \frac{2}{\pi}(\omega - \omega_1) \mathrm{PV} \int_0^\infty \frac{(\varepsilon'(\omega') - 1)(\omega'^2 + \omega\omega_1)}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} \mathrm{d}\omega'$$

Relaciones de KK



$$c_{\mathrm{Hb}} = 28.7\ \mathrm{g/dL}$$

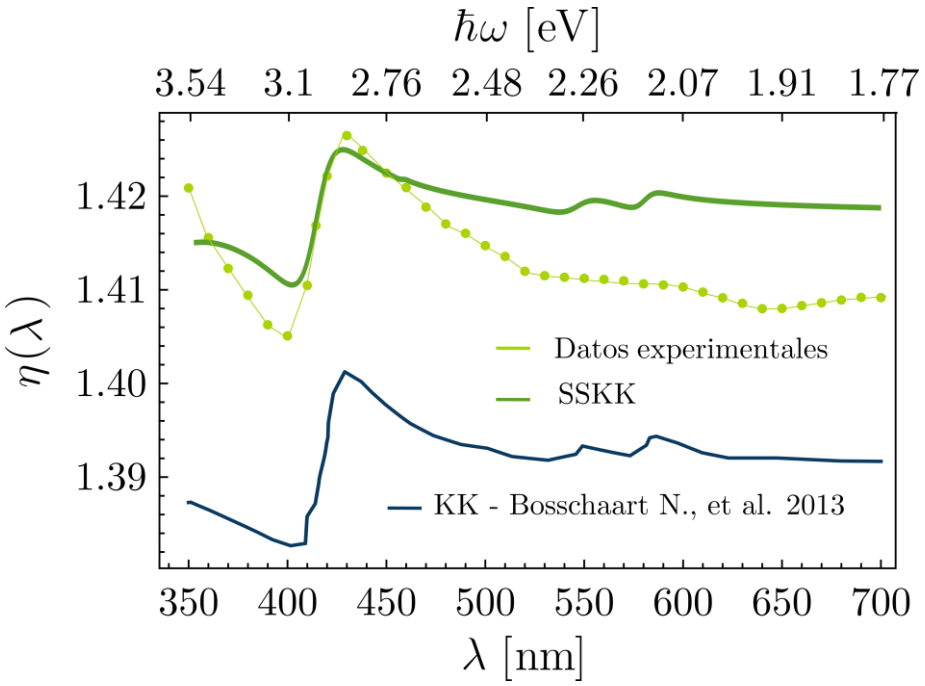
Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005

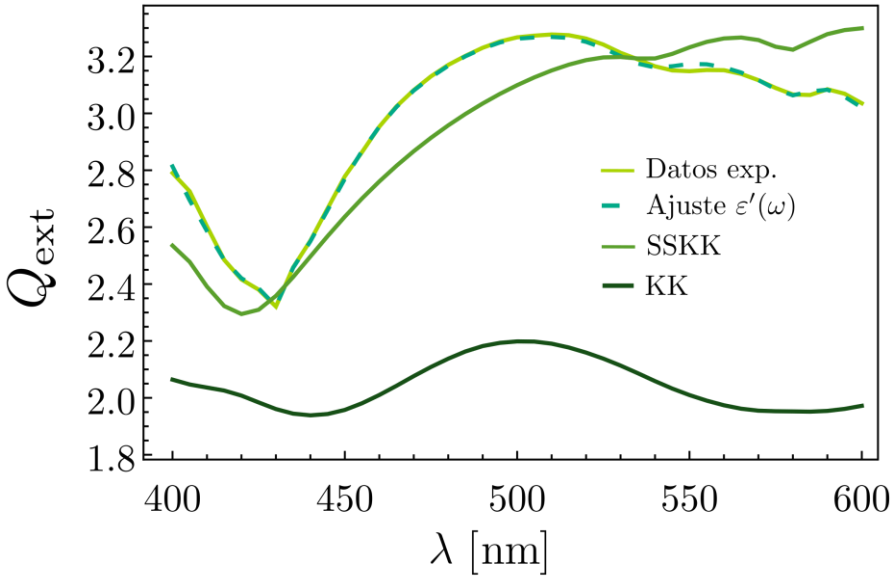
Eficiencia de extinción

Teoría de Mie

Q_{ext} empleando diferentes funciones dieléctricas de la hemoglobina

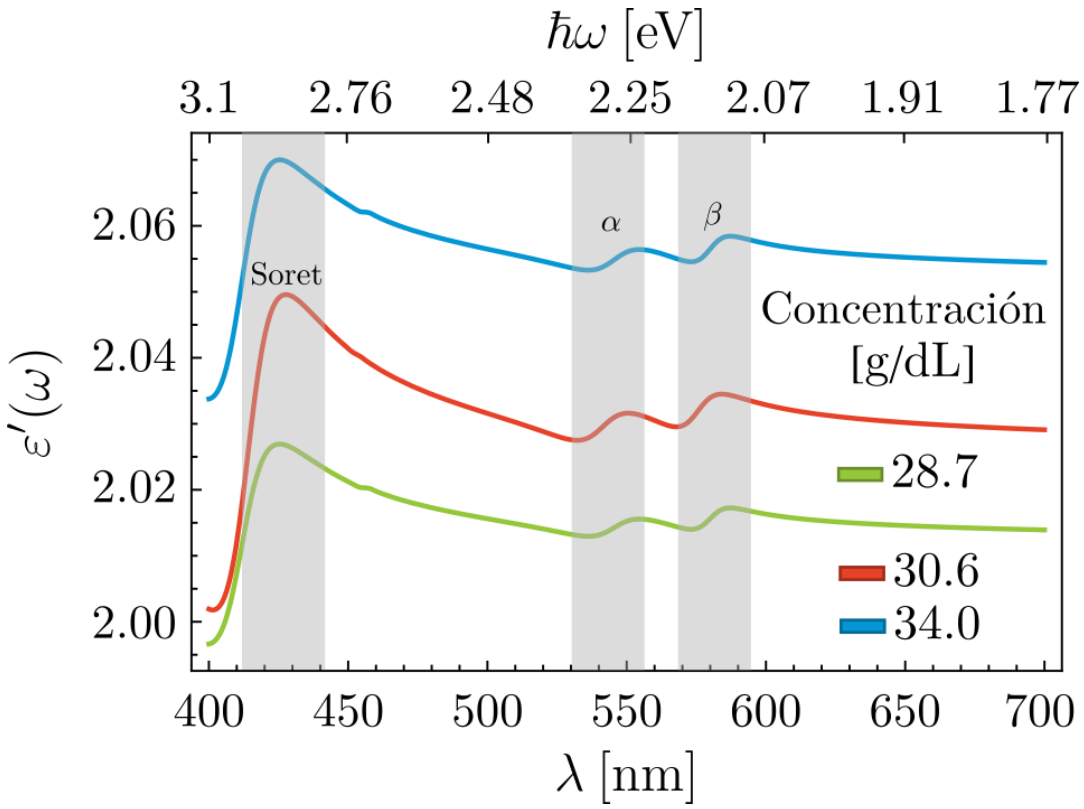
Radio: 2680 nm
Matriz: Plasma ($n_{\text{Plasma}} = 1.823$)
$c_{\text{Hb}} = 28.7$ g/dL

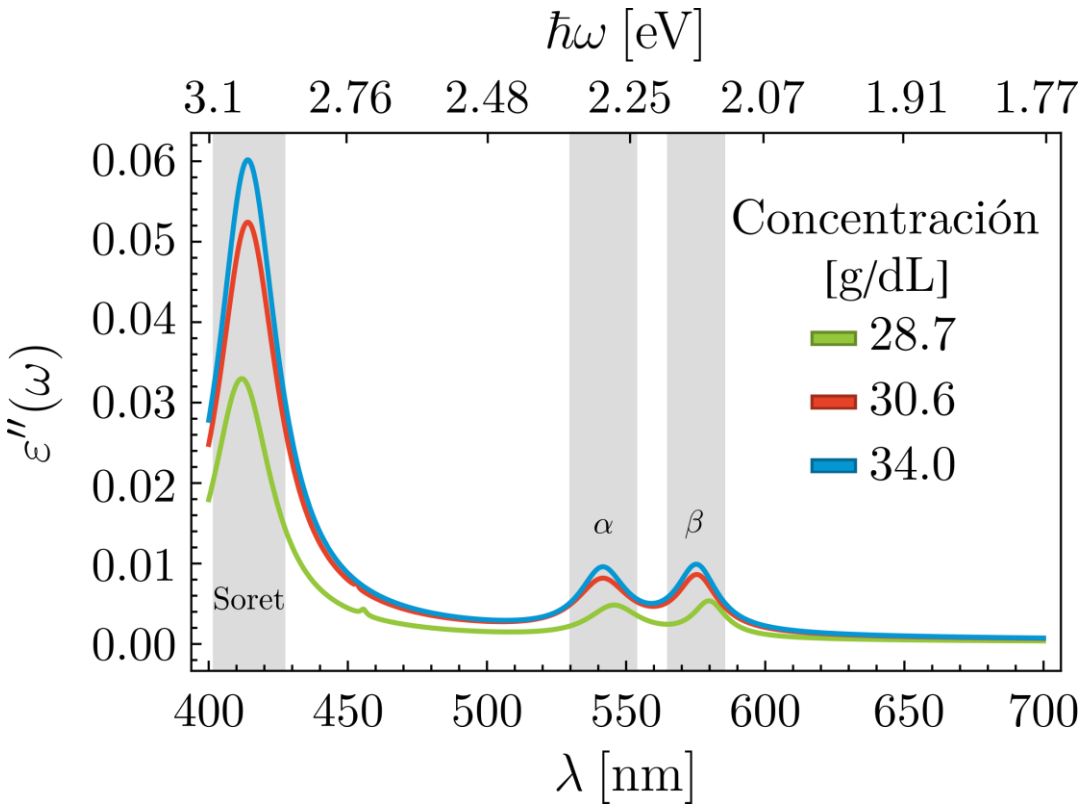




Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005
Bosschaart N., et al. *Lasers in Medical Science*, **29**(2), 2013

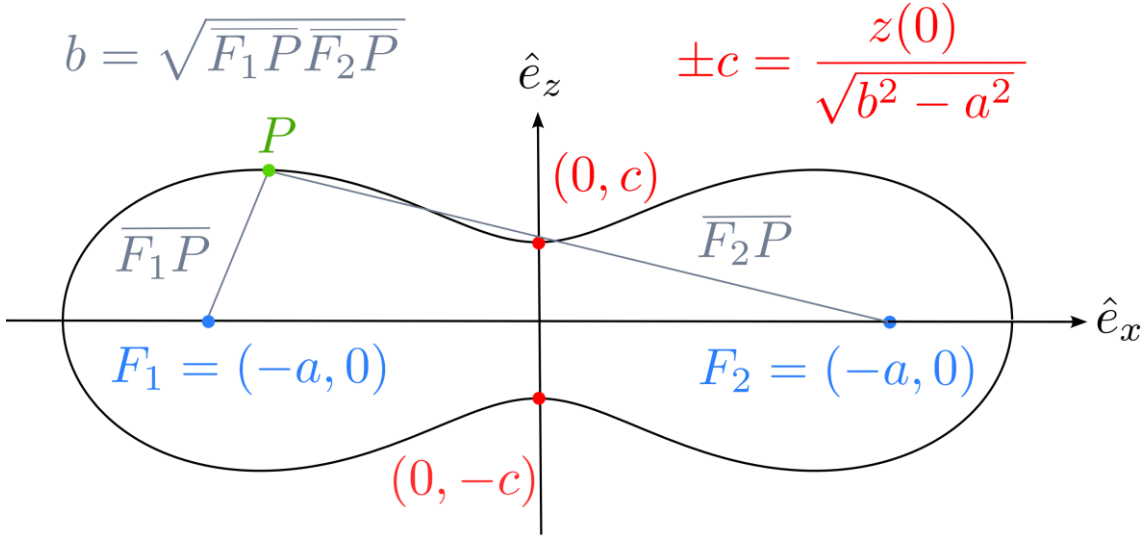
Función dieléctrica

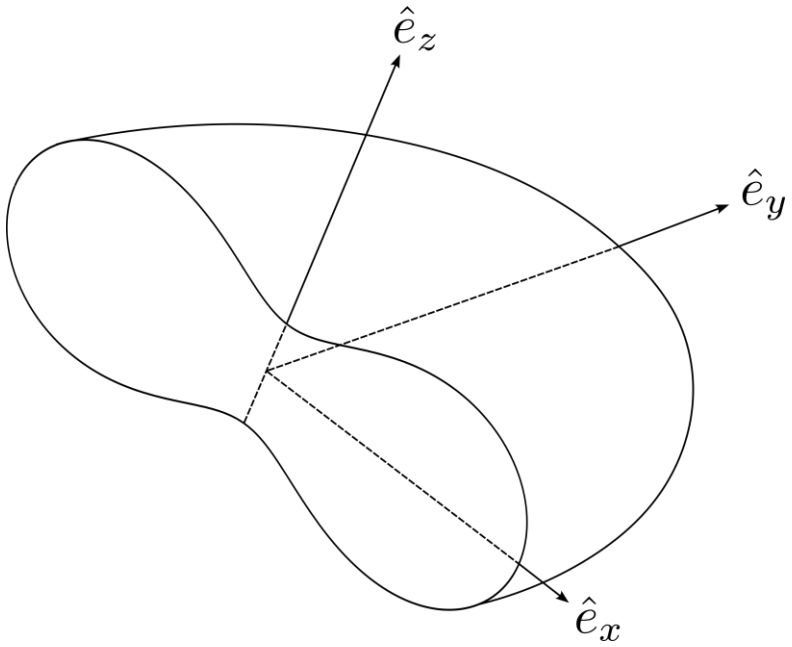




Óvalos de Cassini

$$z = \pm c(-a^2 - x^2 \pm (4x^2a^2 + b^4)^{1/2})^{1/2}$$



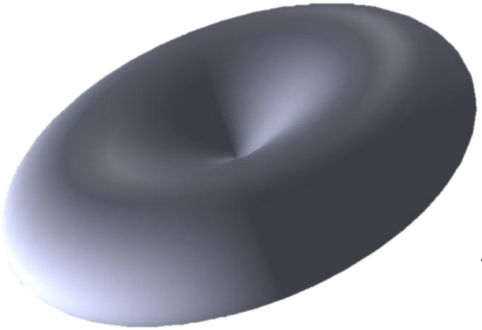


J. Q. Lu, et al. *Journal of Biomedical Optics*, **10**(2), 2005

SolidWorks

Eritrocitos

Eritrocito sano



$$a = 2.6\,\mu\text{m}$$
$$b = 2.7\,\mu\text{m}$$
$$c = 0.8\,\mu\text{m}$$
$$\text{Radio} = 3.76\,\mu\text{m}$$
$$\text{Volumen} = 79.2\,\mu\text{m}^3$$

Microcito

$$a = 2.4\,\mu\text{m}$$
$$b = 2.7\,\mu\text{m}$$
$$c = 0.5\,\mu\text{m}$$
$$\text{Radio} = 3.62\,\mu\text{m}$$
$$\text{Volumen} = 70.40\,\mu\text{m}^3$$

Macrocito

$$a = 2.8\,\mu\text{m}$$
$$b = 3.2\,\mu\text{m}$$
$$c = 0.5\,\mu\text{m}$$
$$\text{Radio} = 4.25\,\mu\text{m}$$
$$\text{Volumen} = 104.51\,\mu\text{m}^3$$

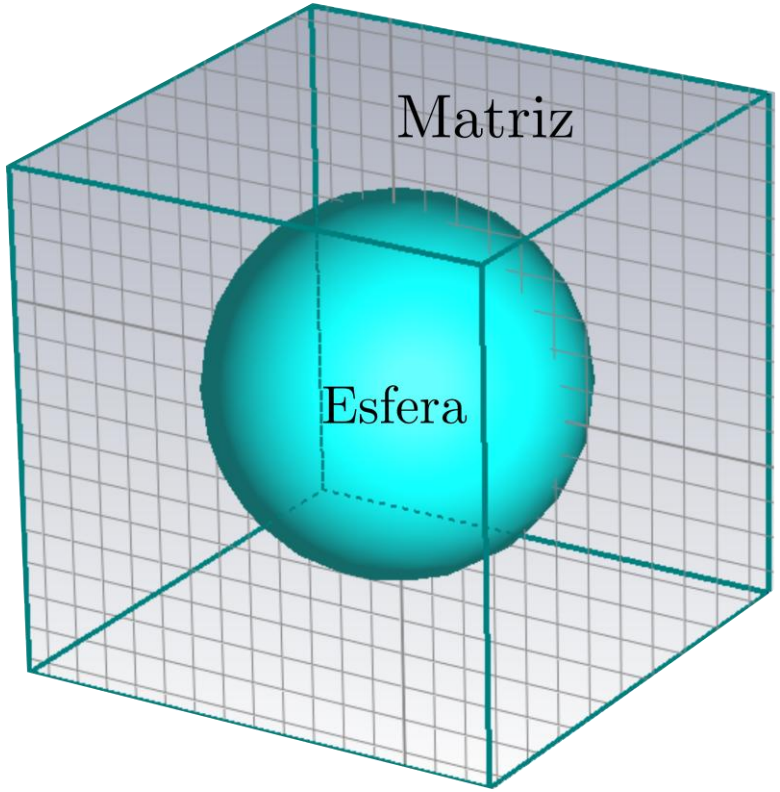
Esferocito

$$\text{Radio} = 2.68\,\mu\text{m}$$
$$\text{Volumen} = 80.63\,\mu\text{m}^3$$

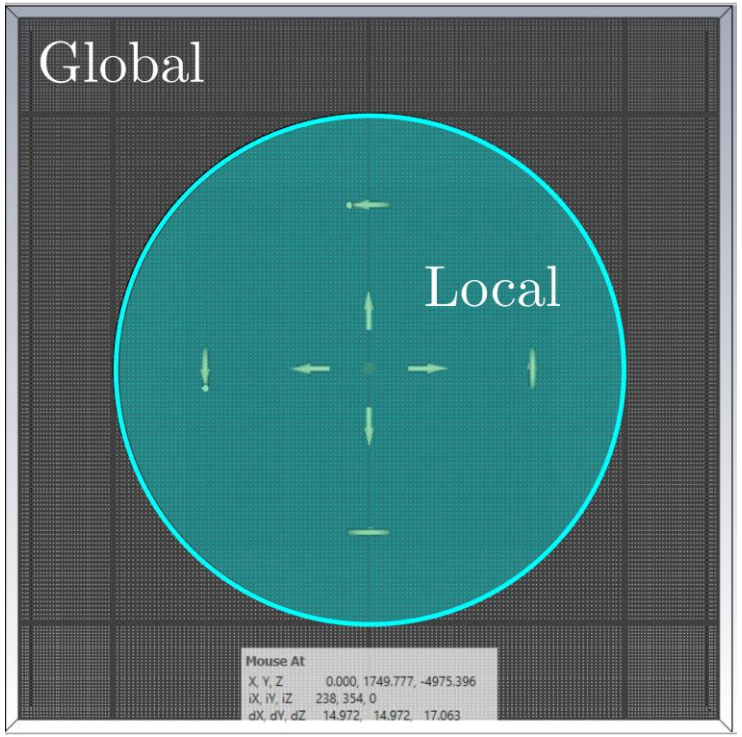
J. Q. Lu, et al. *Journal of Biomedical Optics*, **10**(2), 2005

CST Studio vs Mie

Convergencia



Mallado



Parámetros

- Tamaño de la matriz
- Mallado global
- Mallado local
  - Superficie
  - Volumen

Muchas gracias